

# hexatic graphene

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## 1 Introduction

some relevant papers

Melting of 2D Crystalline Silver Iodide [1]:no evidence has previously been found for it in covalently bonded 2D crystals. Here, 2D crystals of silver iodide embedded in multi-layer graphene are studied with in situ scanning transmission electron microscopy (STEM).

Phase transitions in 2D materials [2]: Some discussion of confined phase transition. no KTHNY or hexatic was mentioned.

The hexatic-solid transition is continuous while, surprisingly, the liquid-hexatic transition is of first order[3]

The relative grain orientations in polycrystalline graphene are normally distributed, with a standard deviation  $\sigma$  in the range of  $4^\circ$  to  $9^\circ$ , based on the full width at half maximum of the etched holes alignment distribution [4].

In classical KTHNY theory, the randomly distributed dislocations are responsible for hexatic phase formation [5][add more].

high energy fore defects migration <https://pubs.acs.org/doi/full/10.1021/acs.jpcc.9b10622>

the local direction field  $\psi_6$ [5][add more]:

$$\psi_6(\vec{r}) = \frac{1}{N_{NN}} \sum_{j \in NN} e^{i6\theta_j},$$

where the sum runs over the  $N_{NN}$  nearest neighbors of atom at position  $\vec{r}$  and  $\theta_j$  is the angle between the bond vector and a fixed global reference axis.

## 2 Polycrystalline Model

We consider a simple polycrystalline model designed to mimic the physical process of random grain nucleation followed by isotropic growth until impingement. Mathematically, the final tiling of the surface by ideal crystallites is described by a two-dimensional Voronoi tessellation constructed from a uniform random distribution of grain centers. The phase of a given structure is characterized by the asymptotic behavior of the bond-orientational correlation function,  $\lim_{r \rightarrow \infty} G_6(r)$ .

Assuming that boundary effects are localized within a characteristic length scale  $l_b \ll \rho^{-1/2}$ , where  $\rho$  denotes the density of grains, the bond-orientational correlation between two random points  $\vec{r}_1$  and  $\vec{r}_2$  located in different crystallites can be approximated as

$$G_6(\vec{r}_2 - \vec{r}_1) = e^{i6\Delta\theta(\vec{r}_1, \vec{r}_2)},$$

where  $\Delta\theta(\vec{r}_1, \vec{r}_2)$  is the relative misorientation between the corresponding grains. The expected value of  $G_6$  therefore depends on the statistical distribution of  $\Delta\theta$  across the polycrystalline network.

### 2.0.1 Ideal Crystallites

In a physical system, the distribution of grain orientations is governed by two competing interactions:

(1) *Interaction with the substrate* [4, 6], which energetically favors global alignment. For small misorientation angles  $\theta$ , the average alignment energy per grain is given by

$$E_a(\theta) = \frac{\epsilon}{\rho} \sum_i \theta_i^2, \quad (1)$$

where  $\epsilon$  is a material-dependent constant, and  $\rho$  is the density of grains.

(2) *Interaction across grain boundaries*, which favors local alignment between neighboring grains. For misorientation angles  $\Delta\theta_{i,j}$  between grain  $i$  and its neighboring grains  $j$ , the interaction energy is expressed according to Read-Shockley model [7]

$$E_g = \frac{\gamma}{3\sqrt{\rho}} \sum_i \sum_{j \in NN_i} |\Delta\theta_{i,j}| (A - \log |\Delta\theta_{i,j}|) \quad (2)$$

where  $\gamma_1, \gamma_2$  are some constants. The average length of a grain boundary segment between two neighboring crystallites is taken as  $L = \frac{2}{3}\rho^{-1/2}$  [8].

this approach gives the total energy of a system:

$$E_{tot} = E_a + E_g$$

The probability of a specific misorientation angle follows the Boltzmann distribution:

$$p(\vec{\theta}) \sim \exp\left(-\frac{E_{tot}(\vec{\theta})}{k_B T}\right).$$

In the limit of large grains ( $\sqrt{\rho} \ll \epsilon/(6\gamma)$ ),  $E_a \gg E_g$ , and global alignment with substrate dominates grains boundaries energy, and the misorientation distribution becomes distance-independent. This results in a crystalline-like orientational correlation function,

$$G_6(r) \propto \exp\left(-18\rho \frac{k_B T}{\epsilon}\right),$$

In contrast, for small grains,...

### 2.0.2 Grain Boundaries (not sure if we need this)

In the previous discussion, we completely neglected the internal structure of grain boundaries. This simplification is justified if the distortions introduced by the boundaries are confined to a finite region around them. In that case, the boundaries merely renormalize the asymptotic behavior of correlation functions without altering the qualitative nature of the phase.

However, the actual spatial extent of this distortion region remains unclear. To quantify it, we model grain boundaries as periodic arrays of dislocations [9]. The bond-angle (lattice rotation) field induced by dislocations with Burgers vectors  $\vec{b}(\vec{r}')$  located at positions  $\vec{r}'$  is given by [5]:

$$\theta(\vec{r}) = \frac{-1}{2\pi} \sum_{\vec{r}'} \frac{\vec{b}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2}$$

## 3 Simulations details

The initial centers of crystallites were randomly distributed within a  $60 \times 60 \text{ nm}^2$  square region, with periodic boundary conditions applied every 20 nm, corresponding to a grain density of 90 grains per  $\mu\text{m}^2$ . Grain geometries were constructed via Voronoi tessellation of the crystallite centers and filled with perfect graphitic lattices, each rotated by a random angle drawn from a normal distribution with standard deviation  $\sigma_a$ .

To optimize the grain boundaries, the central  $25 \times 25 \text{ nm}^2$  region was annealed using molecular dynamics (MD) with the AIREBO force field [10], as implemented in the LAMMPS simulation package [11]. A timestep of  $5 \times 10^{-16} \text{ s}$  was used. The system, initially confined to a two-dimensional plane, was first energy-minimized, then rapidly heated to 5000 K using a Langevin thermostat with a damping parameter of  $10^{-12} \text{ s}$  over 10,000 MD steps. It was subsequently cooled to 0 K over 20,000 MD steps and finally re-optimized at zero temperature.

## Appendix A

?? The estimation of correlation function will be done in analogy with XY model [12]. The Hamiltonian of the system is:

$$\mathcal{H} = \frac{\epsilon}{\rho} \sum_i \theta_i^2 + \frac{\gamma}{3\sqrt{\rho}} \sum_i \sum_{j \in NN_i} |\Delta\theta_{i,j}| (A - \log |\Delta\theta_{i,j}|)$$

## Appendix B

?? The bond-angle (lattice rotation) field induced by dislocations with Burgers vectors  $\vec{b}(\vec{r}')$  located at positions  $\vec{r}'$  is given by [5]:

$$\theta(\vec{r}) = \frac{-1}{2\pi} \sum_{\vec{r}'} \frac{\vec{b}(\vec{r}') \cdot (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^2}$$

In simplified case, we put them in line along  $y$  direction, which gives

$$\theta(\vec{r}) = \frac{-b}{2\pi} \sum_{n=-\infty}^{\infty} \frac{x}{x^2 + (y - nd)^2}$$

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